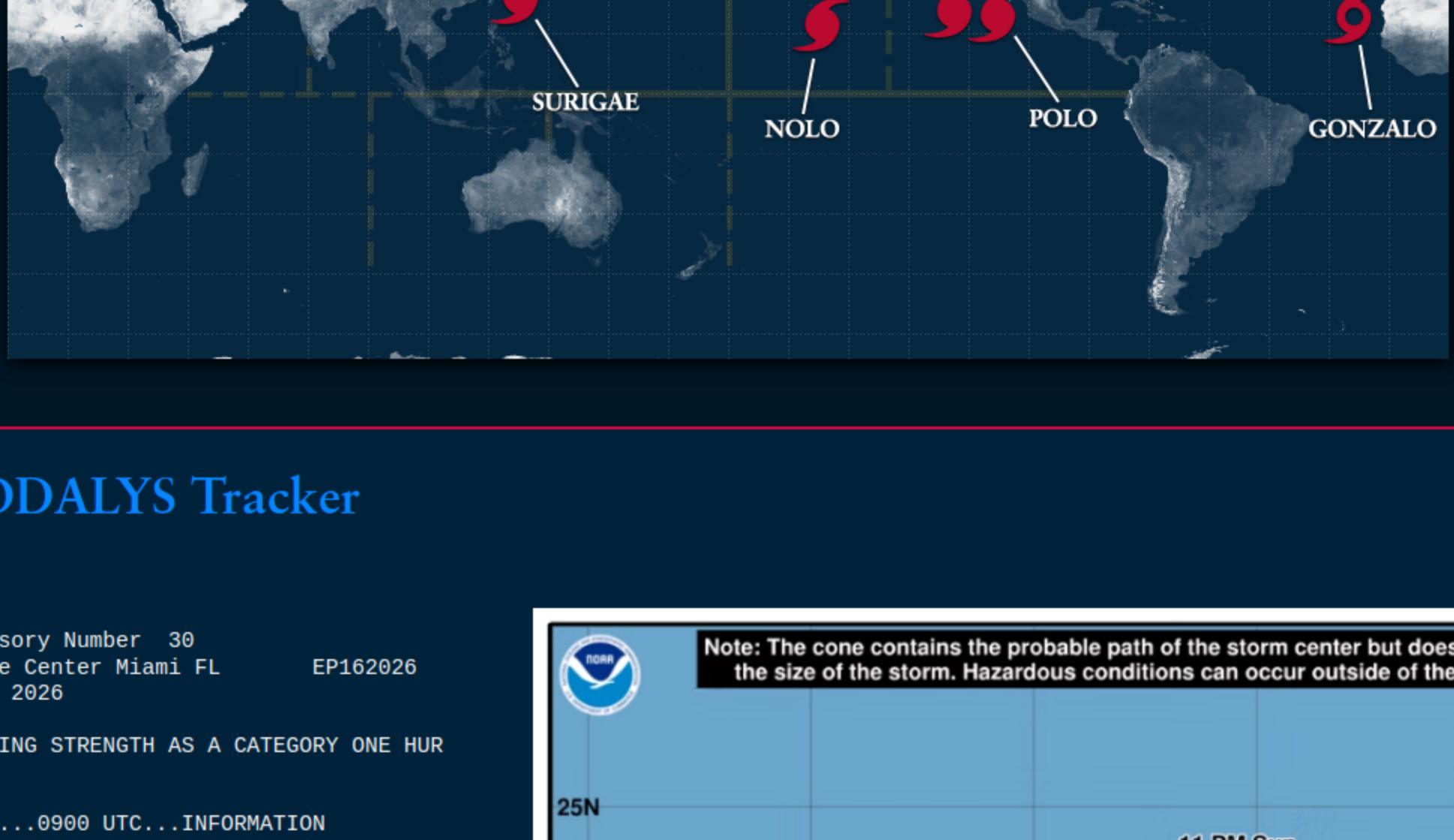


STORMDOM

Global Storm Tracker

Home ▾ Indian Ocean ▾ West Pacific ▾ South Pacific ▾ Central Pacific ▾ East Pacific ▾ Atlantic



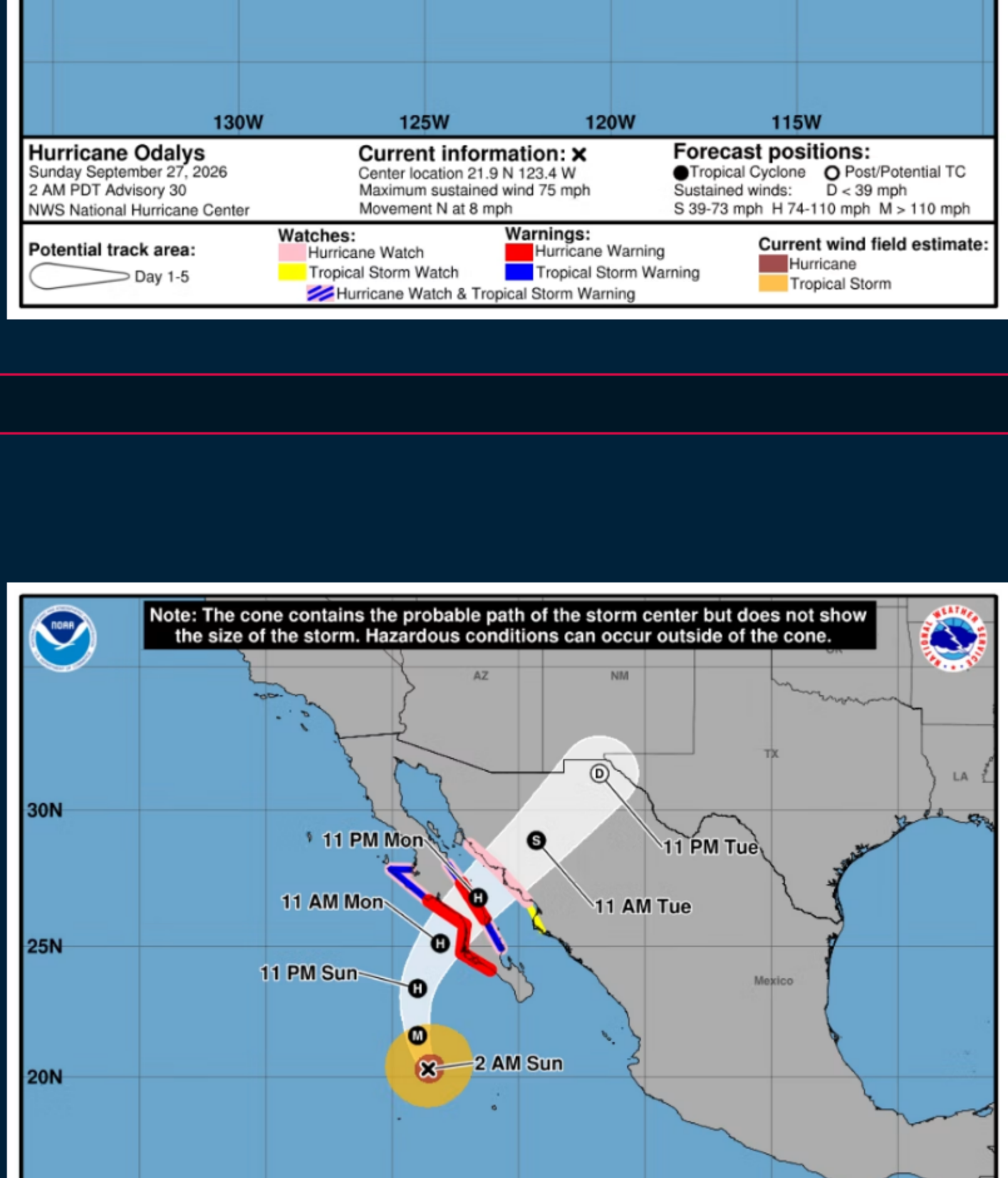
Hurricane ODALYS Tracker

Hurricane Odalys Advisory Number 39
NWS National Hurricane Center Miami FL
200 AM PDT Sun Sep 27 2026

EP162026

...ODALYS RAPIDLY LOSING STRENGTH AS A CATEGORY ONE HUR

SUMMARY OF 200 AM PDT...0900 UTC...INFORMATION
LOCATION...21.9N 123.4W
ABOUT 865 MI...1390 KM W OF THE SOUTHERN TIP OF BAJA CA
MAXIMUM SUSTAINED WINDS...75 MPH...120 KM/H
PRESENT MOVEMENT...N OR S DEGREES AT 8 MPH...13 KM/H
MINIMUM CENTRAL PRESSURE...985 MB...29.69 INCHES



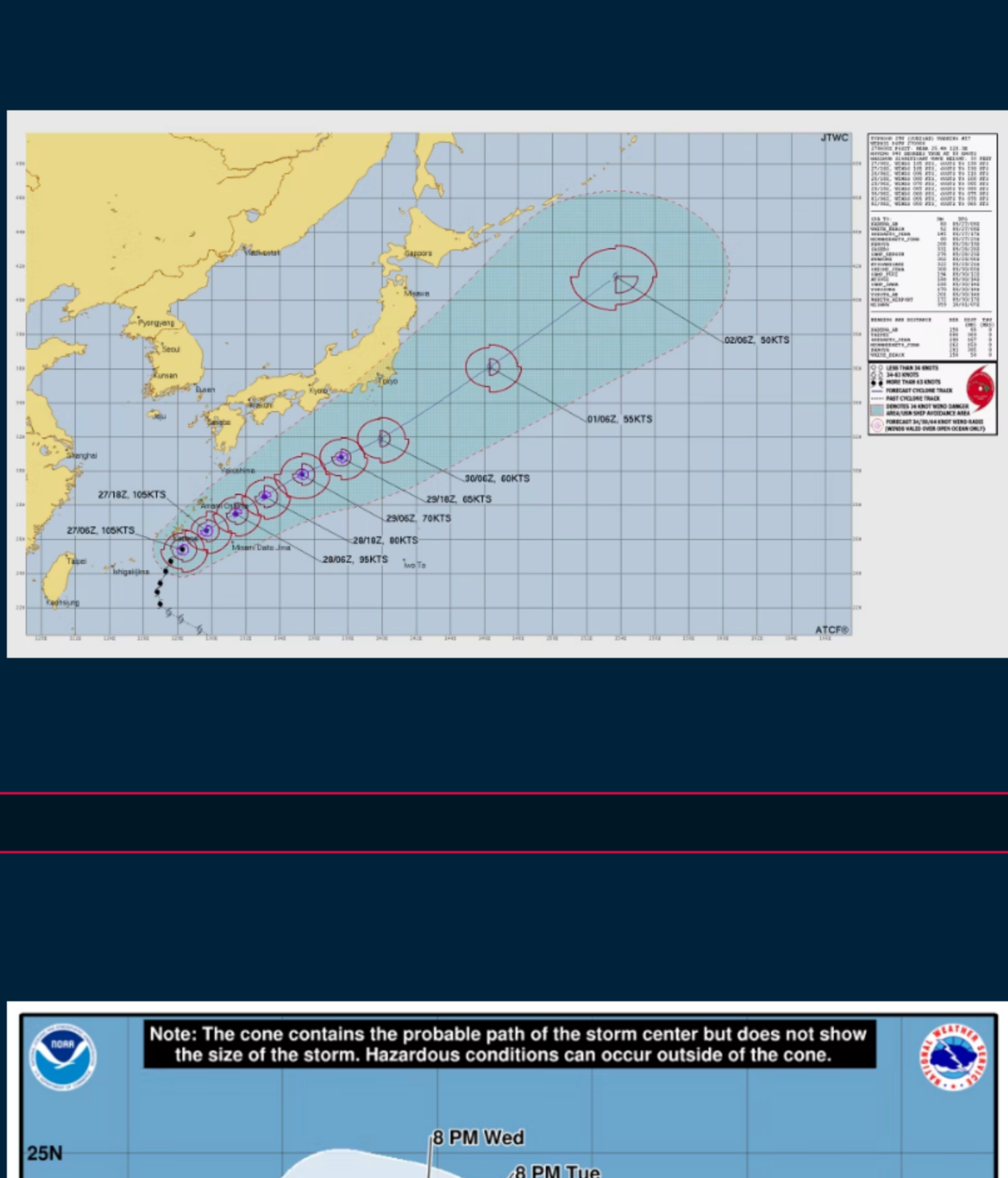
Hurricane POLO Tracker

Hurricane Polo Advisory Number 27
NWS National Hurricane Center Miami FL
200 AM MST Sun Sep 27 2026

EP172026

...NEW WATCHES AND WARNINGS ISSUED FOR PORTIONS OF BAJA
AND MAINLAND NORTHWESTERN MEXICO...
...PREPARATIONS TO PROTECT LIFE AND PROPERTY SHOULD BE
COMPLETION IN BAJA CALIFORNIA SUR...

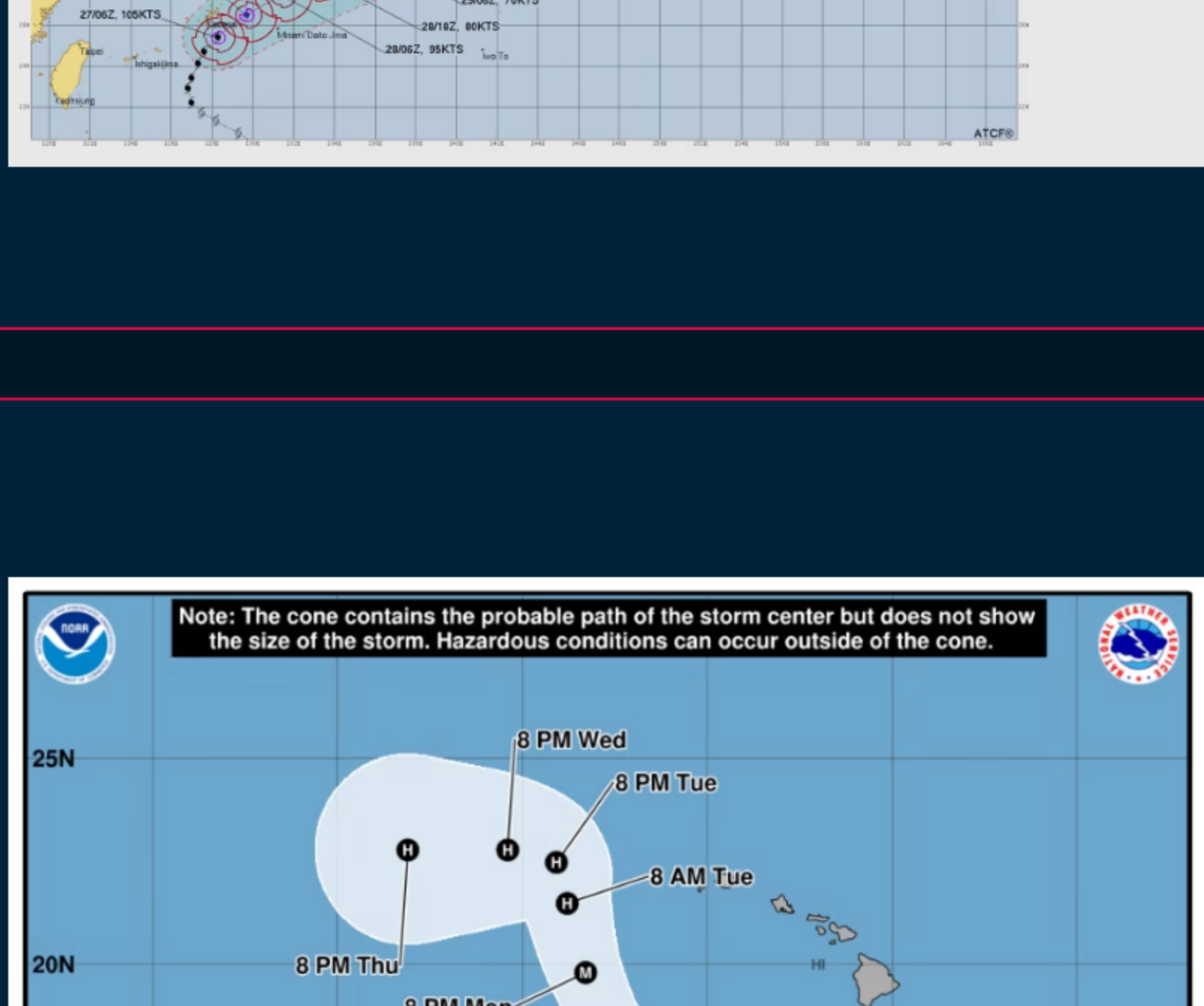
SUMMARY OF 200 AM MST...0900 UTC...INFORMATION
LOCATION...25.3N 113.6W
ABOUT 300 MI...480 KM SW OF CABO SAN LUCAS MEXICO
ABOUT 320 MI...520 KM SW OF CABO SAN LAZARO MEXICO
MAXIMUM SUSTAINED WINDS...120 MPH...195 KM/H
PRESENT MOVEMENT...NNW OR 330 DEGREES AT 10 MPH...17 KM
MINIMUM CENTRAL PRESSURE...952 MB...28.12 INCHES



Typhoon SURIGAE Tracker

1. TYPHOON 25W (SURIGAE) WARNING NR 017
01 ACTIVE TROPICAL CYCLONE IN NORTHWESTPAC
MAX SUSTAINED WINDS BASED ON ONE-MINUTE AVERAGE
WIND RADII VALID OVER OPEN WATER ONLY

WARNING POSITION:
270600Z --N NEAR 25.4N 128.3E
MOVEMENT PAST SIX HOURS - 040 DEGREES AT 09 KTS
POSITION ACQUATE TO WITHIN 010 NM
POSITION BASED ON EYE FIXED BY SATELLITE
PRESENT WIND DISTRIBUTION:
MAX SUSTAINED WINDS 105 KT, GUSTS 130 KT
WIND RADII VALID OVER OPEN WATER ONLY
RADIUS OF 064 KT WINDS - 820 NM NORTHEAST QUADRANT
820 NM SOUTHEAST QUADRANT
820 NM SOUTHWEST QUADRANT
815 NM NORTHWEST QUADRANT
RADIUS OF 050 KT WINDS - 845 NM NORTHEAST QUADRANT
845 NM NORTHEAST QUADRANT
845 NM SOUTHEAST QUADRANT
835 NM SOUTHWEST QUADRANT
RADIUS OF 034 KT WINDS - 840 NM NORTHEAST QUADRANT
870 NM NORTHEAST QUADRANT
880 NM SOUTHEAST QUADRANT
870 NM SOUTHWEST QUADRANT
860 NM NORTHWEST QUADRANT
REPEAT POSIT: 25.4N 128.3E



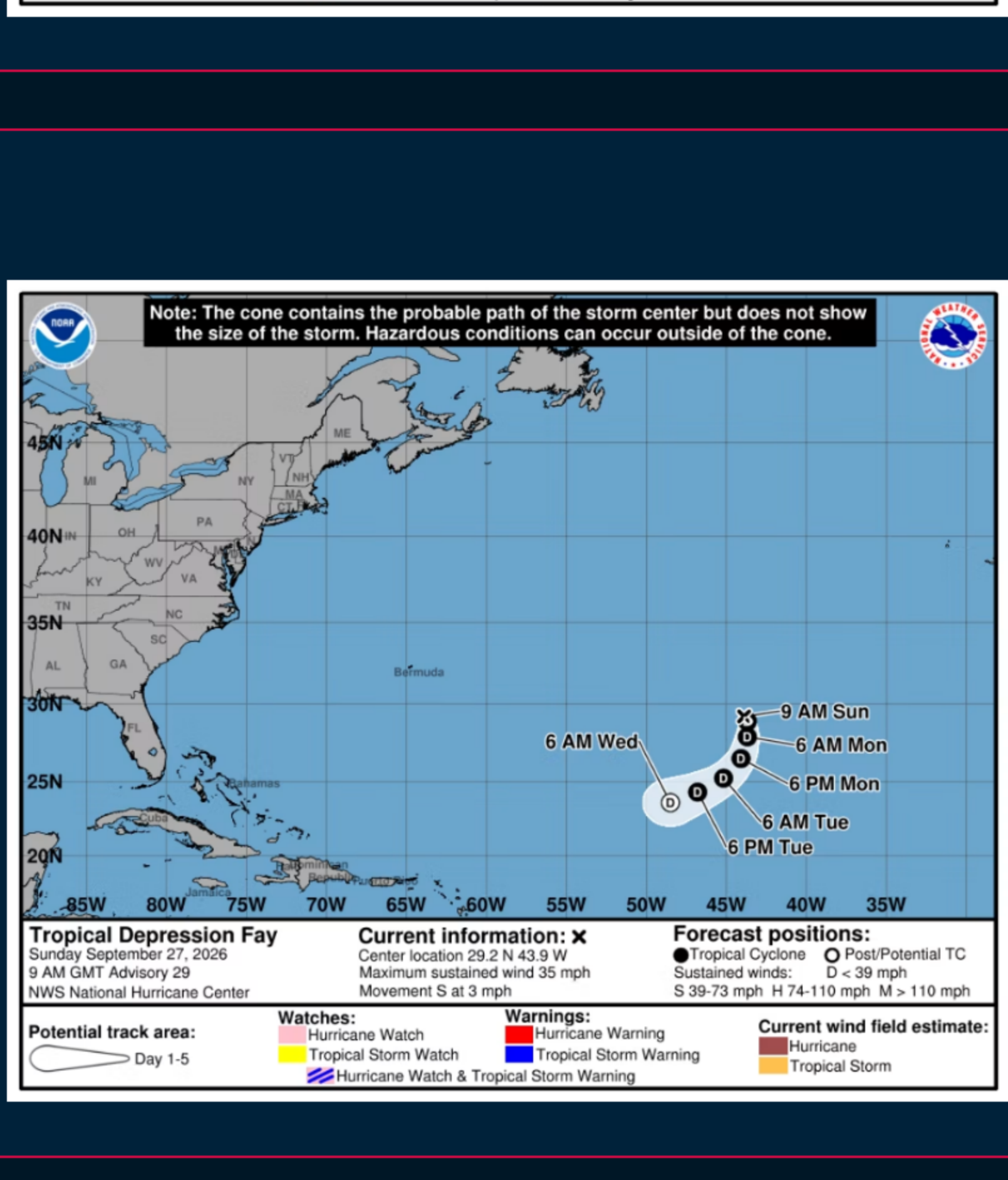
Hurricane NOLO Tracker

Hurricane Nolo Advisory Number 27
NWS National Hurricane Center Honolulu HI
1100 PM HST Sat Sep 26 2026

EP15

...NOLO CONTINUES MOVING WESTWARD WHILE GRADUALLY STRENGTHENING
...GUSTY WINDS, HEAVY RAIN, AND LARGE WAVES CONTINUE FO
OF THE HAWAIIAN ISLANDS...

SUMMARY OF 1100 PM HST...0900 UTC...INFORMATION
LOCATION...16.5N 157.3W
ABOUT 200 MI...325 KM SSW OF SOUTH POINT HAWAII
ABOUT 1145 MI...1840 KM SSW OF HONOLULU HAWAII
MAXIMUM SUSTAINED WINDS...35 MPH...55 KM/H
PRESENT MOVEMENT...W OR 270 DEGREES AT 9 MPH...15 KM/H
MINIMUM CENTRAL PRESSURE...967 MB...28.56 INCHES



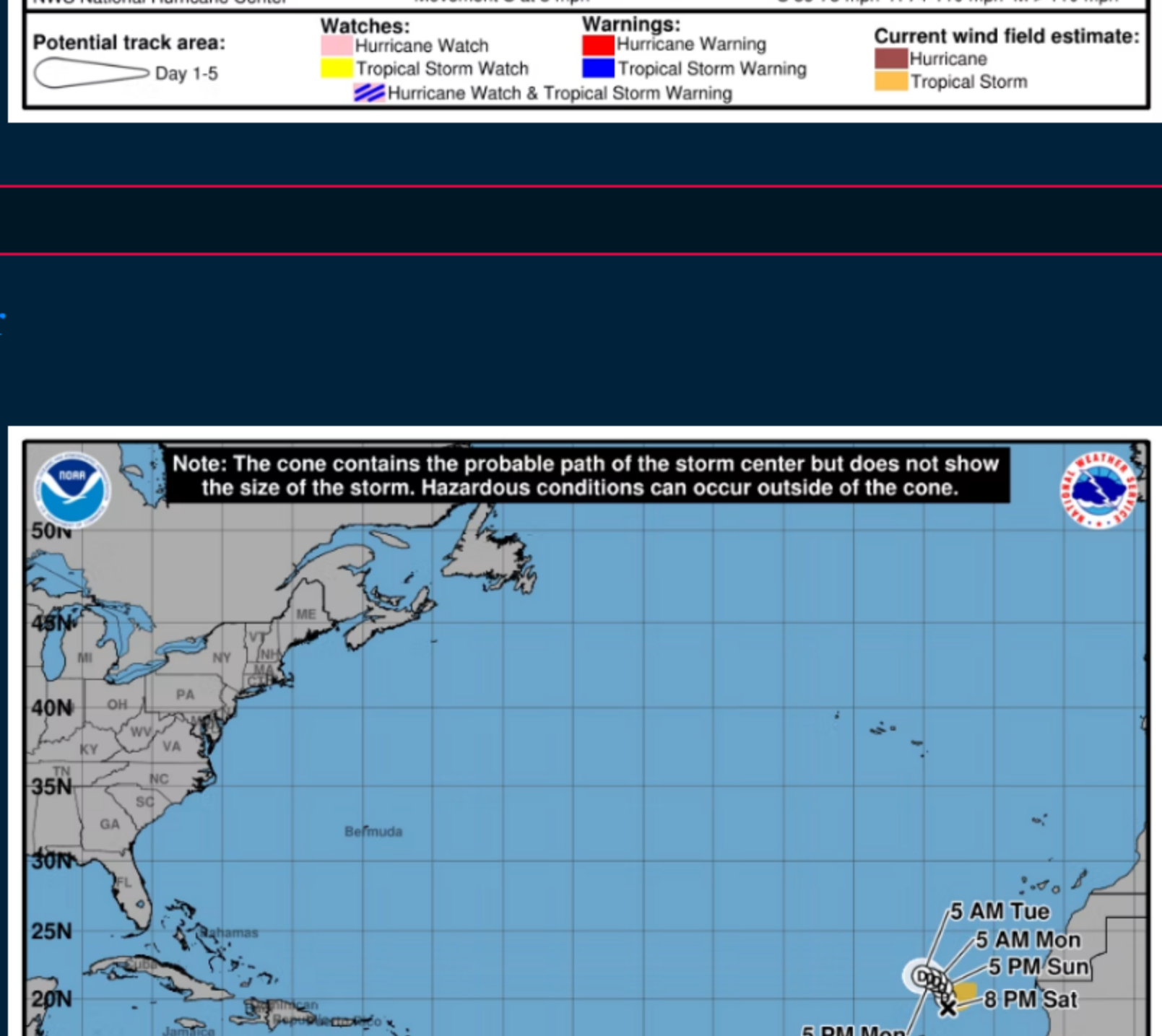
Tropical Depression FAY Tracker

Tropical Depression Fay Advisory Number 29
NWS National Hurricane Center Miami FL
900 AM GMT Sun Sep 27 2026

AL862026

...FAY HOLDING STEADY AS IT MOVES SOUTHWARD...

SUMMARY OF 900 AM GMT...0900 UTC...INFORMATION
LOCATION...29.2N 43.9W
ABOUT 1145 MI...1840 KM WSW OF THE AZORES
MAXIMUM SUSTAINED WINDS...35 MPH...55 KM/H
PRESENT MOVEMENT...S OR 175 DEGREES AT 3 MPH...6 KM/H
MINIMUM CENTRAL PRESSURE...1010 MB...29.83 INCHES



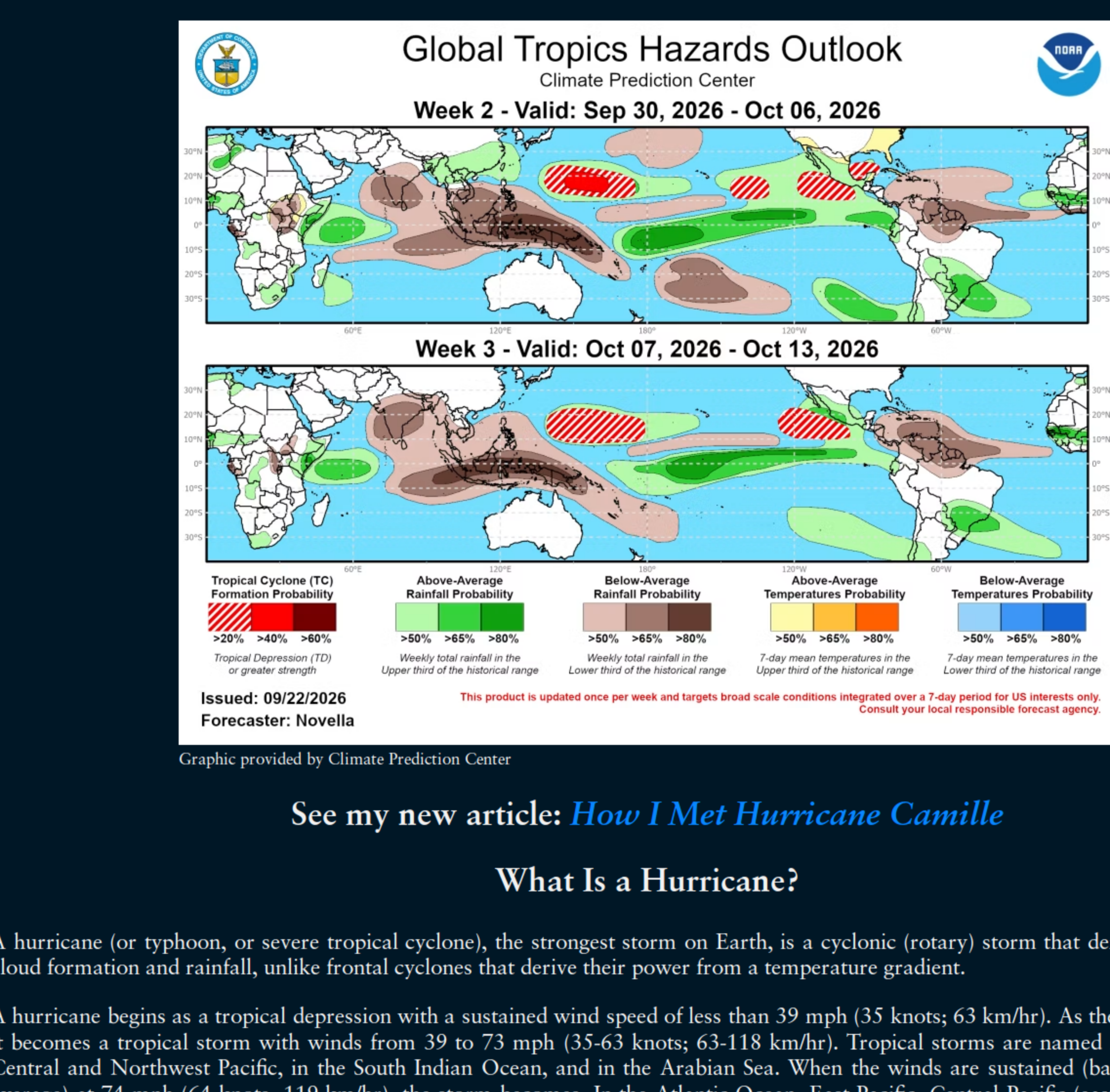
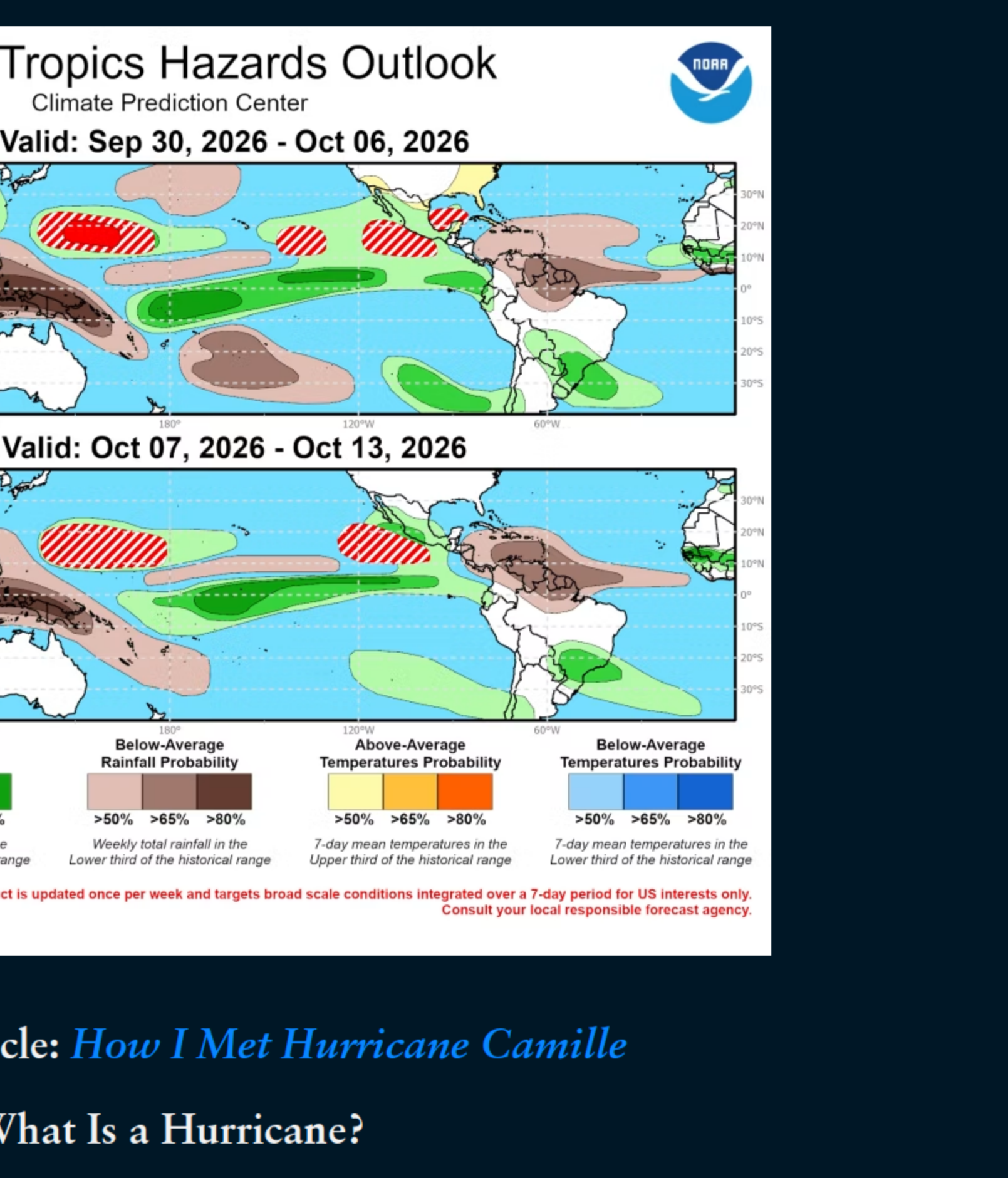
Tropical Storm GONZALO Tracker

Post-Tropical Cyclone Gonzalo Advisory Number 8
NWS National Hurricane Center Miami FL
800 PM CDT Sat Sep 26 2026

AL872026

...GONZALO NO LONGER A TROPICAL CYCLONE...
...THIS IS THE FINAL ADVISORY...

SUMMARY OF 800 PM CDT...2100 UTC...INFORMATION
LOCATION...19.3N 22.3W
ABOUT 180 MI...290 KM N OF SAL CABO VERDE
MAXIMUM SUSTAINED WINDS...40 MPH...65 KM/H
PRESENT MOVEMENT...NNW OR 345 DEGREES AT 7 MPH...11 KM/H
MINIMUM CENTRAL PRESSURE...1005 MB...29.68 INCHES



See my new article: [How I Met Hurricane Camille](#)

What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the strongest storm on Earth, is a cyclonic (rotary) storm that derives its energy from cloud formation and rainfall, unlike frontal cyclones that derive their power from a temperature gradient.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (35 knots; 63 km/hr). As the system strengthens, it becomes a tropical storm with winds from 39 to 73 mph (35-63 knots; 63-118 km/hr). Tropical storms are named in the Atlantic, East, Central and North Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots; 119 km/hr), the storm becomes: In the Atlantic Ocean, East Pacific, Central Pacific (east of the International Dateline) and Southeast Pacific (east of 160°E) a Hurricane; in the Northwest Pacific (west of the International Dateline) a Typhoon; in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) a Severe Tropical Cyclone; in the North Indian Ocean a Severe Cyclonic Storm; and in the Southwest Indian Ocean (west of 90°E) a Tropical Cyclone.

The Saffir-Simpson Hurricane Scale

Category 1 – 64-82 knots (74-95 mph; 119-153 km/h). Damage is limited to foliage, signage, unanchored boats and mobile homes. There is no significant damage to buildings. The main threat to life and property may be flooding from heavy rains.

Category 2 – 83-95 knots (96-110 mph; 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers damaged by storm surge. Some trees blown down, more extensive limb damage.

Category 3 – 96-112 knots (111-129 mph; 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes are completely destroyed. Roof damage is common. Storm surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

Category 4 – 113-136 knots (130-156 mph; 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extreme storm surge damage and flooding. Severe coastal erosion, with permanent changes to the coastal landscape not unheard of. Hurricane force winds extend well inland.

Category 5 – 137+ knots (157+ mph; 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/hr) is called a Super Typhoon.

When the wave passes over warmer waters (SSTs), convection and resulting rain greater rainfall is concomitant with falling surface pressures. By the time a wave is 1008mb, it is likely that the northeast wind has closed off to a southwest wind wave. The forward motion of the wave completes the closure on the northern side center, and a tropical depression has formed.

We often hear that a tropical depression has formed, but conditions are unfavorable for development. There are two conditions that must be present for the tropical development: warm SSTs (above 79° Fahrenheit/26° Celsius) and low vertical shear. The storm derives its power from the warm waters below. In addition, a strong anticyclone

Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by how quickly the water depth increases offshore. In deep-water environments, such as the Hawaiian islands, storm surge will be enhanced by the rapidly decreasing ocean depth as the wind-driven surge approaches the coast. The peak storm surge is on the right-front quadrant (left-front in the Southern Hemisphere) of the eyewall at landfall, where on-shore winds are the strongest, and at the leading edge of the eyewall. Contrary to a popular myth, the storm surge is entirely wind-driven water—not it is not caused by the low pressure of the eye. Another factor in the severity of the storm surge is tide. Obviously, an 18-foot storm surge at high tide is that much worse than an 18-foot surge at low tide.

Tropical Cyclone Formation

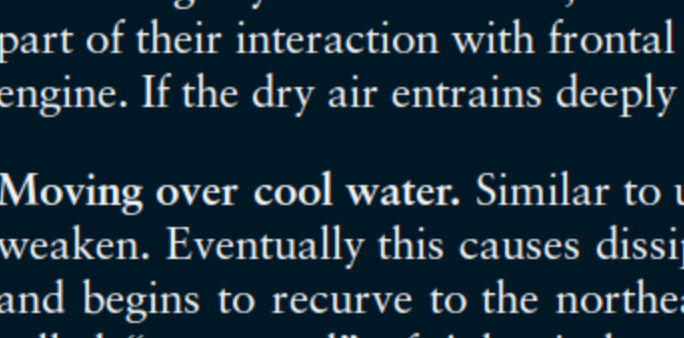
By Jonathan Edwards

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

This occurs when, in the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation after about the middle of May.

A series of low-pressure ripples develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west-northwest, or even northwest, after crossing 45° or 50° W longitude.

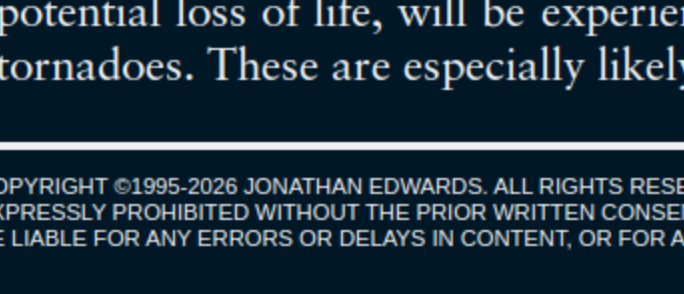
These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a “convergence”, where lines of equal atmospheric pressure are pressed together between the high-pressure ridge to the north and the developing low-pressure system. The divergence that results ahead of the convergence zone gives us a northeasterly wind as the axis of the tropical wave approaches. Gusts up to 25 mph may occur. Sometimes there can be gusts to tropical storm force in stronger waves. There can be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave.



(Graphic by Robert Simmon, NASA GSFC)

When the wave passes over warmer waters (SSTs), convection and resulting rainfall are enhanced. This greater rainfall is concomitant with falling surface pressures. By the time these pressures fall to 1008mb, it is likely that the northeast wind has closed off to a southwest wind along the backside of the wave. The forward motion of the wave completes the closure on the northern side of a broad low-level center, and a tropical depression has formed.

We often hear that a tropical depression has formed, but conditions are unfavorable for further development. There are two conditions that must be present for the tropical depression to continue to develop: warm SSTs (above 79° Fahrenheit/26° Celsius) and low vertical shear. A tropical cyclone derives its power from the warm waters below. In addition, a strong anticyclone directly above the low-level inflow is favorable. As a tropical cyclone is pulling in warm, moist air at the surface, it must also evacuate this inbound flow aloft. This occurs in the upper levels of the atmosphere, where high pressure facilitates the development of the cyclone by evacuating the flow from the lower levels of the cyclone. Every powerful hurricane has an equally powerful high pressure system over it. At the surface, the air spirals inward in a counter-clockwise direction, rises in the developing center, and spirals out at the top in a clockwise direction. In the Southern Hemisphere, it is reversed: clockwise inbound, counterclockwise outbound.



(Image courtesy NOAA)

Eye formation begins when a tropical storm reaches approximately 65mph, provided conditions are favorable for strengthening to continue. The eyewall begins to make its appearance, usually on the eastern (Northern Hemisphere) edge of the center. As the system becomes better organized and the smaller, more defined center. The inflow is spiralling in ever faster as it is evacuated up through the developing eyewall and out by the high pressure outflow structure. The eye begins to appear as a clear spot in the center, as the air here is sinking. The eyewall creates subsidence that not only helps clear the eye, but can also produce a feature known as the “moat”, which is an area of relatively weak convection outside the core of the tropical cyclone. The combination of eye and moat can make the storm’s core look like a doughnut. The eye can have a diameter anywhere from 10 to 40 miles across.

Tropical cyclones can exhibit a great deal of durability provided that the upper level support remains and the southerly (Northern Hemisphere), or tropical, inflow is present. Here are some examples: In 1998, Hurricane Mitch developed into a supermassive Category 5 hurricane, nevertheless weakening began when the center moved to a position directly north of Honduras, cutting off the southerly inflow, even though the eye was still over a hundred miles offshore. Later, Mitch maintained its mid-level core against all odds over the mountains of Central America because he was able to advect moisture from the East Pacific. Reintensification to tropical storm strength was almost immediate after reaching the Bay of Campeche. In 1918 Gilbert hit the Yucatan near Cancun as a Category 5 hurricane. The Yucatan peninsula, though flat, extended far enough south through the critical southerly inflow zone that Gilbert never recovered, even after moving over the open waters of the western Gulf of Mexico. Contrast this with the inflow Gulf coast and numerous examples from Camille to Elena and beyond, where the proximity of the eye to land is not necessarily a weakening factor, although the extensive shelf water can sometimes be an inhibiting factor, if a slow-moving storm generates enough upwelling. A recent Southern Hemisphere example of this is Severe Tropical Cyclone Monica, which developed into a “super cyclone” just north of Australia. The northerly (equatorward) inflow over the Arafura Sea fueled this system.

What are the factors that contribute to the decay of a tropical cyclone? They are Upwelling, Entraining dry air, Moving over cool waters, Exposure to upper-level westerlies, and finally Landfall.

Upwelling. When a hurricane stalls, its movement is has fallen below 5mph, or its movement is erratic over a small area, the wave action caused by the strong surface winds churns the ocean surface and produces upwelling. This has the effect of cooling the temperature of the sea surface over an area 200 to 300 miles across. The result is weakening. It is possible for a hurricane to stall in one area long enough that it dissipates. In 2004, Hurricane Frances stalled off the southeast Florida coast long enough that the core of the hurricane collapsed. Frances eventually made landfall as a Category 1 hurricane, instead of the earlier expected Category 3.

Entraining dry air. Sometimes, during the peak season, when tropical cyclones approach continental land masses, they may entrain dry air as part of their interaction with frontal troughs that carry cool, dry air behind them. The dry air kills off the convection that drive the hurricane’s engine. If the dry air entrains deeply enough, the result can be dissipation.

Moving over cool water. Similar to upwelling, when a tropical cyclone moves over cool water (below 77° Fahrenheit/25° Celsius), it begins to weaken. Eventually this causes dissipation, particularly in the East Pacific. In the Atlantic, if the storm is caught in the mid-latitude westerlies and begins to recurve to the northeast, it generally becomes an extratropical storm by the time it has reached about 45° W. This is the so-called “graveyard” of Atlantic hurricanes. The storm becomes extratropical when it has transitioned to a cold-core, baroclinically driven system, and eventually becomes a gale in the north Atlantic, or is absorbed by a large gale. There have been times when a hurricane passes north of the Azores and hits the British Isles as a Force 8 or stronger gale, having maintained a recognizable inner core.

Exposure to upper-level westerlies. Frequently, the ridge of high pressure that keeps the hurricane heading toward the west erodes. In the Atlantic Ocean, this may result in strong upper-level westerlies diving down and impinging on the northern edge of the hurricane’s upper-level support structure. Poleward outflow is restricted. A hurricane can still maintain itself when outflow is restricted in one quadrant. If outflow becomes too restricted, resulting shear can disrupted the inner core.

After landfall. Tropical cyclones cannot survive over land, once their access to warm SST is removed. A powerful hurricane, such as Hugo, which hit Charleston, S.C., in 1989 can project life-threatening hurricane force winds over two hundred miles inland. As the storm progresses inland, it can dump a huge amount of rain—sometimes measured in feet. The storm may evolve into a frontal cyclone that continues to cause widespread damage. The best example of this is Hurricane Camille in 1969—the strongest hurricane ever to make landfall on the continental United States with winds sustained at 190 mph and gusts well exceeding 200 mph—which roared up the Mississippi Valley and eventually exited off the East Coast. Camille maintained tropical storm strength as far as Memphis, Tennessee. Most hurricanes will diminish in strength rapidly after landfall, reaching tropical depression strength by 48 to 72 hours. The main threat from the dying storm is from tornadoes and inland flooding. The right-hand quadrant of a hurricane or typhoon (in the Northern Hemisphere, left-hand in the Southern Hemisphere) is most frequently the strong side of the storm. This is because the forward motion is added to the counter-clockwise punch of the storm. Thus, the right-hand side of the hurricane contains the strongest on-shore flow. This is where the maximum storm surge flooding, and the greatest potential loss of life, will be experienced. After landfall, the circulation of the storm over land causes a great deal of turbulence, which results in tornadoes. These are especially likely in the forward right-hand side of the storm’s path.

COPYRIGHT 2025-2026 JONATHAN EDWARDS. ALL RIGHTS RESERVED. PORTIONS COPYRIGHT THEIR RESPECTIVE OWNERS. REPUBLICATION OR REDISTRIBUTION OF STORMDOM CONTENT, INCLUDING BY FRAMING OR SIMILAR MEANS, IS EXPRESSLY PROHIBITED WITHOUT THE PRIOR WRITTEN CONSENT OF JONATHAN EDWARDS. THE STORMDOM BANNER AND THE WORDS “GLOBAL STORM TRACKER” ARE A TRADE MARK OF JONATHAN EDWARDS. JONATHAN EDWARDS SHALL NOT BE LIABLE FOR ANY ERRORS OR DELAYS IN CONTENT, OR FOR ANY ACTIONS TAKEN IN RELIANCE THEREON.